7/27/2026

[ShadowFox Data Science Internship]

[Advanced Level Task]

Netflix Movies & TV Shows Data Analysis Using Python]

priyanshu1117r@hotmail.com
[SHADOWFOX DATA SCIENCE INTERNSHIP]

Introduction:

Netflix is one of the world's leading online streaming platforms, offering thousands of movies, TV shows, documentaries, and original content across various countries and languages. With millions of subscribers worldwide, Netflix continuously expands its content library to meet the diverse preferences of its audience.

In this project, Python libraries such as **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn** are used for data cleaning, exploratory data analysis (EDA), and visualization. The project aims to uncover meaningful insights about Netflix's content library through statistical analysis and graphical representations.

Objectives:

The main objectives of this project are:

- To analyze the Netflix Movies and TV Shows dataset.
- To clean and preprocess the dataset for analysis.
- To compare the number of Movies and TV Shows available on Netflix.
- To identify the countries producing the highest amount of content.
- To analyze content release trends over the years.
- To study the distribution of content ratings.
- To identify the most common genres and directors.
- To visualize the data using various charts and graphs.
- To derive meaningful insights that help understand Netflix's content library.

Scope of the Project:

This project focuses on analyzing the Netflix Movies and TV Shows dataset using Python. The analysis includes data cleaning, handling missing values, exploratory data analysis (EDA), and data visualization. Different charts and graphs are used to understand content distribution, release trends, country-wise production, ratings, genres, and other important characteristics of the dataset.

The project is limited to the available dataset and does not include real-time Netflix data. However, the analysis demonstrates practical data science techniques that can be applied to similar real-world datasets. The findings can help understand streaming content patterns and improve decision-making based on historical data.

Research Questions:

The project aims to answer the following research questions:

1. How many Movies and TV Shows are available on Netflix?
2. Which country has produced the highest number of Netflix titles?
3. Which year has the highest number of content releases?
4. Which content ratings are most common?
5. Which directors have directed the highest number of titles?

Dataset Description:

The dataset used in this project is **Netflix Movies and TV Shows Dataset**. It contains information about movies and TV shows available on Netflix.

The dataset includes the following important attributes:

- **show_id** – Unique identifier for each title.
- **type** – Specifies whether the content is a Movie or TV Show.
- **title** – Name of the movie or TV show.
- **director** – Director of the content.
- **cast** – Main actors and actresses.
- **country** – Country where the content was produced.
- **date_added** – Date when the content was added to Netflix.
- **release_year** – Year of release.
- **rating** – Audience rating (TV-MA, PG, TV-14, etc.).
- **duration** – Duration of movies or number of seasons for TV shows.
- **listed_in** – Genre or category of the content.
- **description** – Short summary of the content.

The dataset is suitable for performing exploratory data analysis, visualization, and extracting meaningful insights about Netflix's content library.

Tools and Technologies used:

The following tools and technologies were used to complete this project:

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- MS Word
- GitHub

These tools helped in importing, cleaning, analyzing, visualizing, and documenting the dataset efficiently.

Import Libraries, Data Loading, Cleaning:

```
# Netflix Movies and TV Shows Data Analysis Using Python

## ShadowFox Data Science Internship

### Advanced Level Task

**Submitted By:** Priyanshu Gond
```

Import Libraries:

Code Snippet:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
plt.style.use('ggplot')
```

Explanation:

The project uses Python libraries such as Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for creating visualizations.

Uploading Dataset in Google Colab:

Code Snippet:

```
from google.colab import files
uploaded = files.upload()
```

Output:

Choose Files netflix_titles.csv
netflix_titles.csv(text/csv) - 3399671 bytes, last modified: 7/23/2026 - 100% done
```
Saving netflix_titles.csv to netflix_titles.csv
```

Explanation:

This code imports the files module from Google Colab and uploads a dataset from the local computer. The uploaded file is stored in the uploaded variable for further analysis.

Load Dataset:

Code Snippet:

```
import pandas as pd
df = pd.read_csv("netflix_titles.csv")
df.head()
```

Output:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

Explanation:

The dataset was loaded into a Pandas DataFrame using the read_csv() function. The head () function was used to display the first five records and verify that the dataset was imported successfully.

Load Dataset Last Five Rows:

Code Snippet:

```
df.tail()
```

Output:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...

Explanation:

The dataset information provides details about the number of rows, columns, data types, and missing values. Descriptive statistics help summarize the characteristics of the dataset.

Dataset Information:

Explanation:

The df.info() function displays basic information about the dataset, including the number of rows and columns, column names, data types, and non-null values. It helps in understanding the structure of the dataset before analysis.

Code Snippet:

```
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

Dataset Shape:

Code Snippet:

```
df.shape
```

Output:

```
(8807, 12)
```

Explanation:

The df.shape function displays the total number of rows and columns in the dataset. It helps to understand the size of the dataset before analysis.

Dataset Columns:

Code Snippet:

```
df.columns
```

Output:

```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')
```

Explanation:

The df.columns function displays the names of all columns in the dataset. It helps identify the available features for data analysis.

Missing Values:

Explanation:

The df.isnull().sum() function counts the number of missing (null) values in each column of the dataset. It helps identify incomplete data before preprocessing and analysis.

Code Snippet:

```
df.isnull().sum()
```

Output:

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

Statical Summary:

Code Snippet:

```
df.describe(include="all")
```

Output:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2019.000000	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2021.000000	NaN	NaN	NaN	NaN

Explanation:

The df.describe(include="all") function generates a statistical summary of all columns in the dataset, including both numerical and categorical data. It helps in understanding the overall distribution and characteristics of the data.

Check Duplicates Raws:

Code Snippet:

```
df.duplicated().sum()
```

Output:

```
np.int64(0)
```

Explanation:

The df.duplicated().sum() function counts the total number of duplicate rows in the dataset. It helps identify duplicate records that may need to be removed before data analysis.

Remove Duplicates Raws:

Code Snippet:

```
df.drop_duplicates(inplace=True)
```

Explanation:

The df.drop_duplicates(inplace=True) function removes duplicate rows from the dataset. The inplace=True parameter updates the original dataset without creating a new copy.

Fill Missing Values:

Code Snippet:

```
df["director"] = df["director"].fillna("Unknown")
df["cast"] = df["cast"].fillna("Unknown")
df["country"] = df["country"].fillna("Unknown")
df["rating"] = df["rating"].fillna("Not Rated")
```

Explanation:

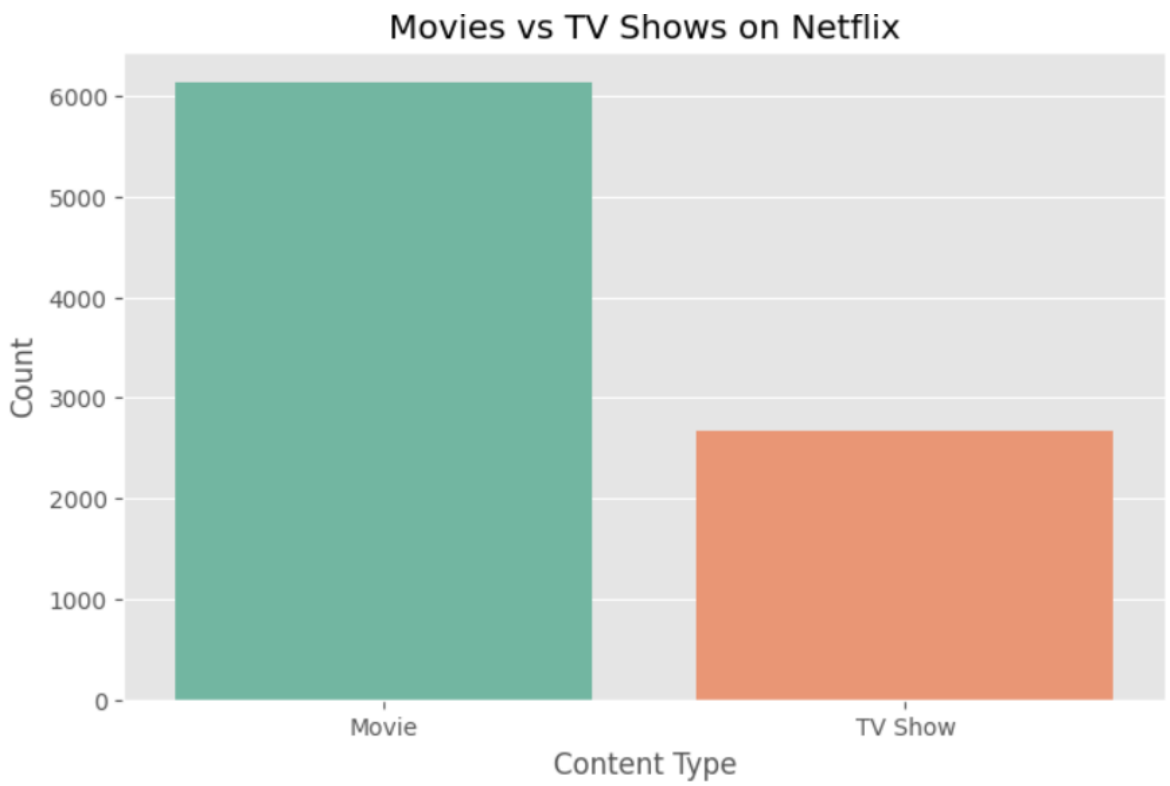
These statements replace missing values in the **director**, **cast**, and **country** columns with **"Unknown"**, and missing values in the **rating** column with **"Not Rated"**. This helps make the dataset complete and ready for analysis.

Movies Vs TV Shows (Count Plot):

Code Snippet:

```
plt.figure(figsize=(8,5))
sns.countplot(x='type', data=df, palette='Set2')
plt.title("Movies vs TV Shows on Netflix")
plt.xlabel("Content Type")
plt.ylabel("Count")
plt.show()
```

Output:



Explanation:

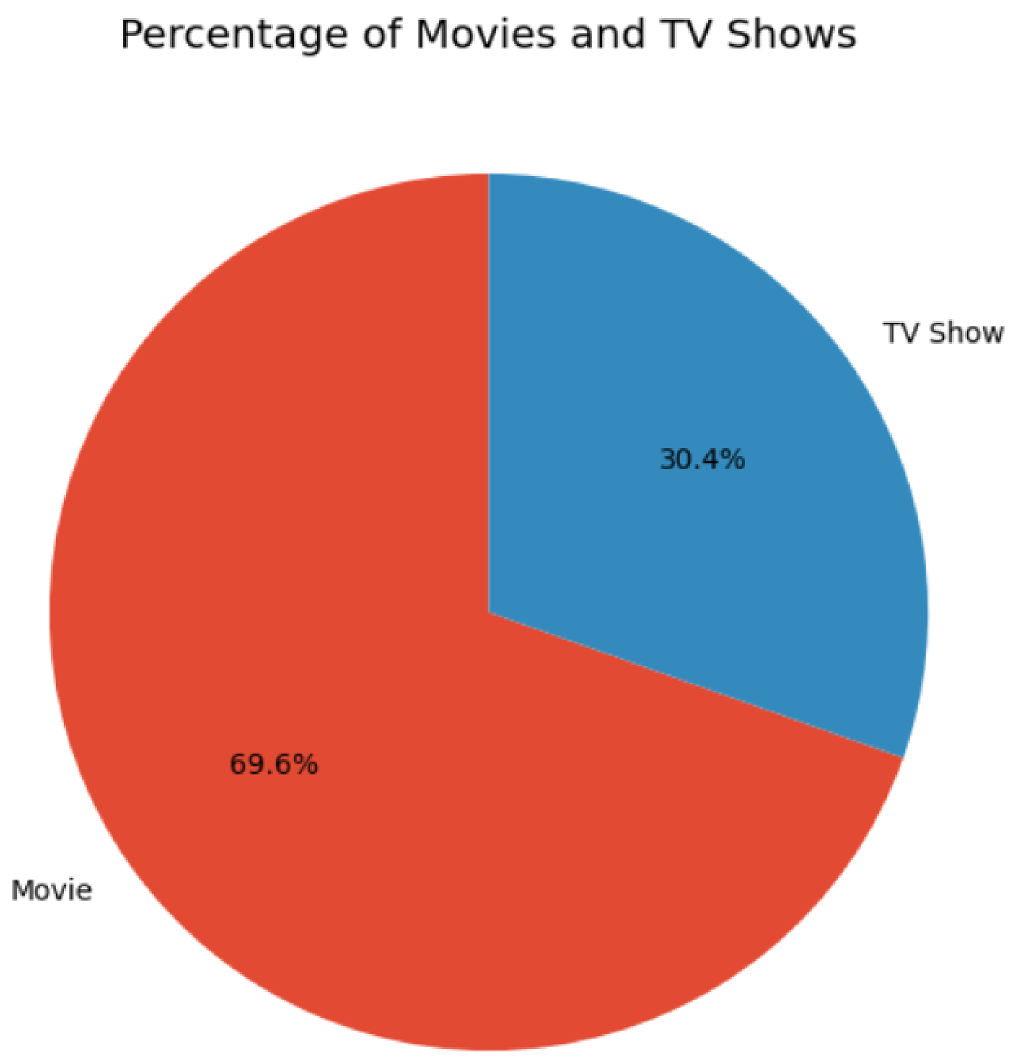
This code creates a **count plot** to show the number of **Movies** and **TV Shows** in the Netflix dataset. It also sets the figure size, chart title, axis labels, and displays the plot for better visualization.

Movies Vs TV Shows (Pie Chart):

Code Snippet:

```
plt.figure(figsize=(7,7))
df['type'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    startangle=90
)
plt.title("Percentage of Movies and TV Shows")
plt.ylabel("")
plt.show()
```

Output:

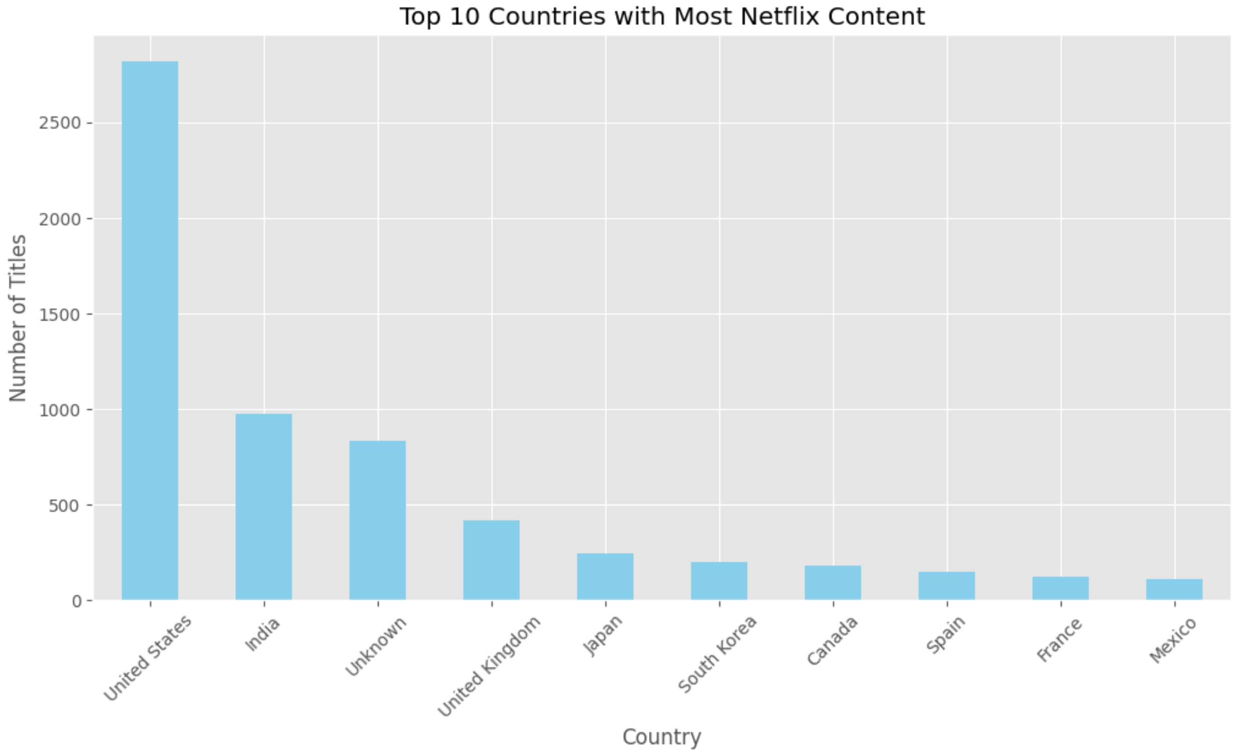


Top 10 Countries (Bar Chart):

Code Display:

```
plt.figure(figsize=(12,6))
df['country'].value_counts().head(10).plot(
    kind='bar',
    color='skyblue'
)
plt.title("Top 10 Countries with Most Netflix Content")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

Output:



Explanation:

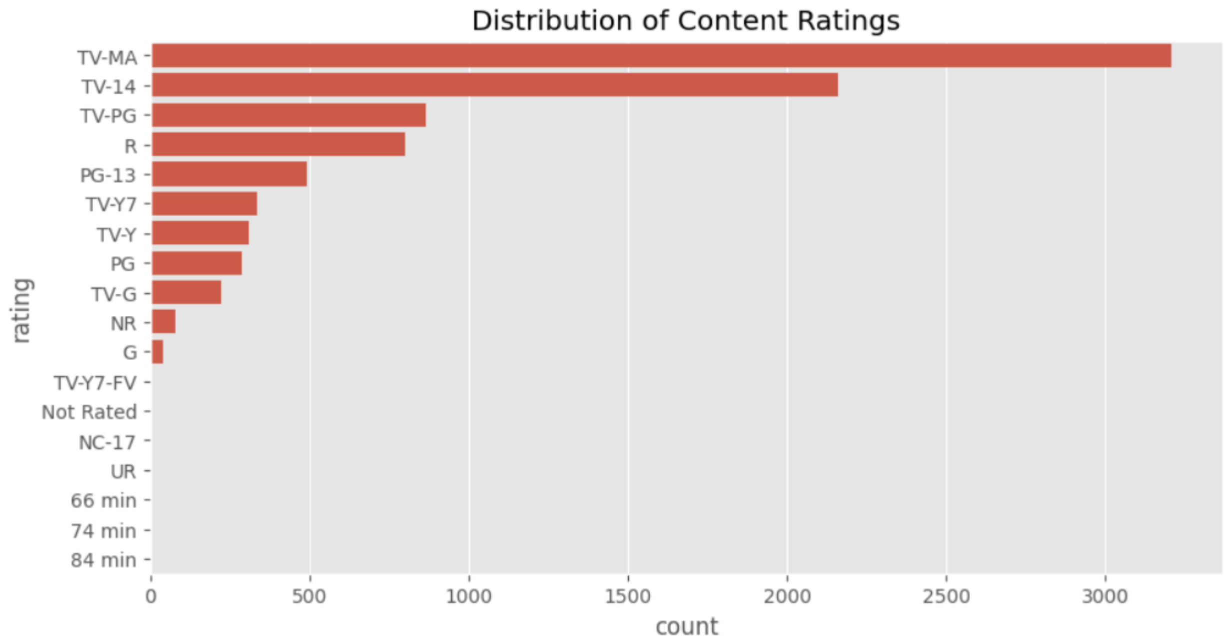
This code creates a **bar chart** showing the **top 10 countries** with the highest number of Netflix titles. It also sets the figure size, chart title, axis labels, rotates the country names for better readability, and displays the plot.

Content Rating Distribution (Count Plot):

Code Snippet:

```
plt.figure(figsize=(10,5))
sns.countplot(
    y='rating',
    data=df,
    order=df['rating'].value_counts().index
)
plt.title("Distribution of Content Ratings")
plt.show()
```

Output:



Explanation:

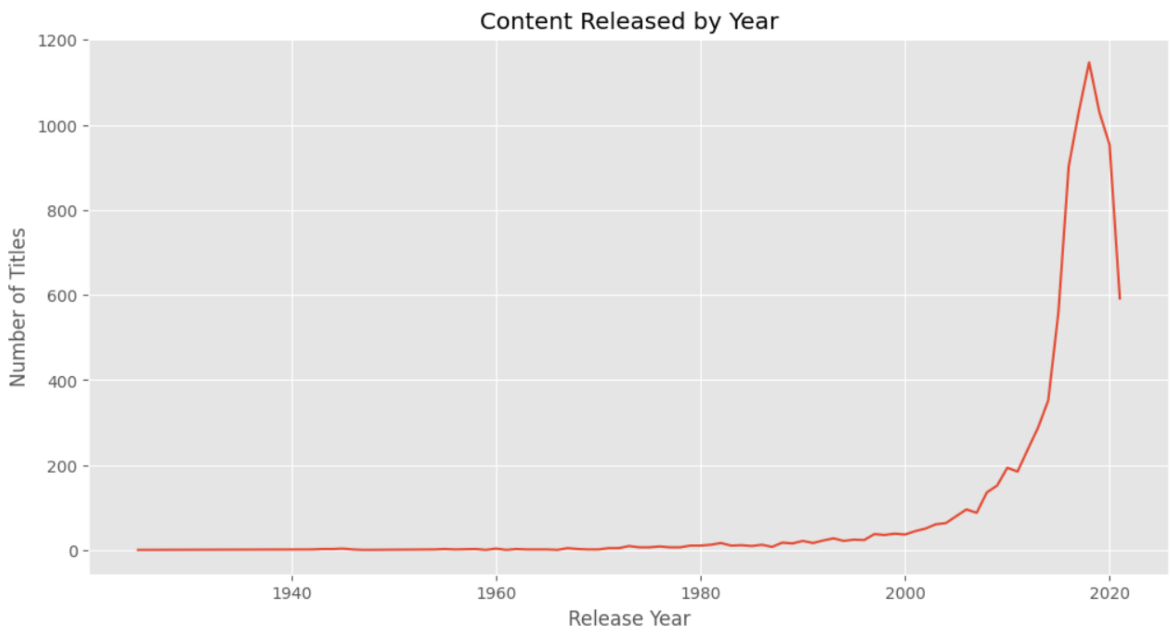
This code creates a **count plot** to show the distribution of different content ratings in the Netflix dataset. The ratings are displayed in descending order based on their frequency, making it easier to compare the number of titles in each rating category.

Content Released by Year (Line Plot):

Code Snippet:

```
plt.figure(figsize=(12,6))
df['release_year'].value_counts().sort_index().plot()
plt.title("Content Released by Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```

Output:



Explanation:

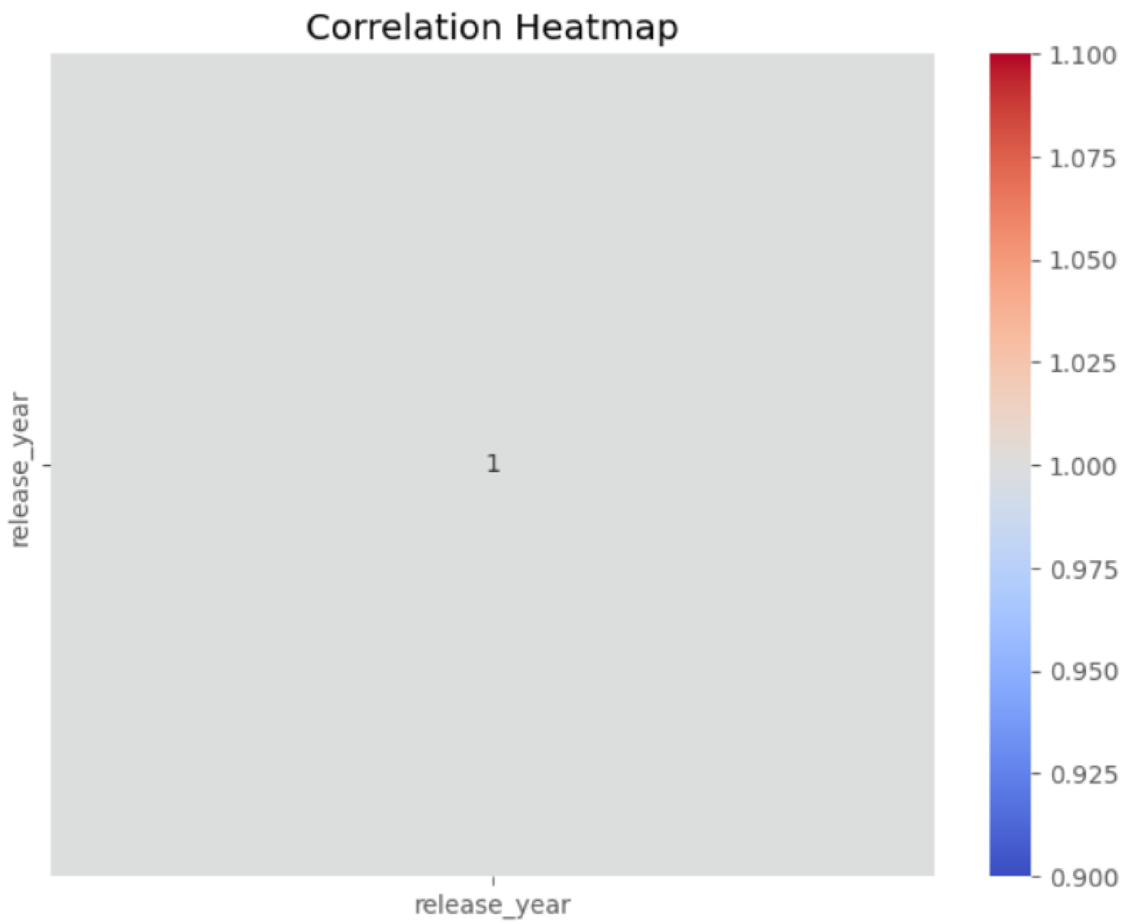
This code creates a **line plot** showing the number of Netflix titles released each year. It helps visualize trends in content releases over time by displaying the count of titles for each release year.

Generating Correlation Heatmap:

Code Snippet:

```
numeric_df = df.select_dtypes(include=['int64','float64'])
plt.figure(figsize=(8,6))
sns.heatmap(
    numeric_df.corr(),
    annot=True,
    cmap='coolwarm'
)
plt.title("Correlation Heatmap")
plt.show()
```

Output:



Explanation:

This code selects all numerical columns from the dataset and creates a **correlation heatmap** to show the relationship between them. The heatmap displays correlation values with annotations, making it easier to identify positive, negative, and strong correlations between variables.

Discussion:

The analysis of the Netflix Movies and TV Shows dataset provides valuable insights into the content available on the platform. The exploratory data analysis (EDA) reveals that Netflix offers a larger number of Movies compared to TV Shows, indicating that movies form a significant portion of its content library.

Country-wise analysis shows that a few countries contribute a major share of Netflix content, with the United States being one of the leading contributors. The release year analysis highlights a significant increase in content production over recent years, reflecting Netflix's rapid expansion and growing investment in original programming.

Conclusion:

This project successfully analyzed the Netflix Movies and TV Shows dataset using Python. The dataset was cleaned, processed, and explored using various data analysis techniques. Multiple visualizations were created to identify trends related to content type, release year, country, ratings, genres, directors, and movie duration.

The analysis showed that Netflix contains a higher number of Movies than TV Shows and that the platform has experienced significant growth over the years. The visualizations also highlighted the most active countries, common content ratings, and popular genres.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn proved to be effective tools for performing data analysis and creating informative visualizations. This project demonstrates the importance of data visualization in understanding large datasets and supporting data-driven decision-making.

Future Scope:

The project can be extended in several ways:

- Develop machine learning models to recommend movies and TV shows.
- Perform sentiment analysis using user reviews.
- Build an interactive dashboard using Streamlit or Power BI.
- Analyze trends across different streaming platforms such as Amazon Prime Video and Disney+.
- Perform predictive analysis to estimate future content growth.
- Integrate live streaming data for real-time analysis.
- Create personalized recommendation systems based on user preferences.

References:

. Netflix Movies and TV Shows Dataset (Kaggle)

. Python Official Documentation

. Pandas Documentation

. NumPy Documentation

. Matplotlib Documentation

. Seaborn Documentation

. Google Colab Documentation

. GitHub Documentation